System Design Workshop: Multi-model Agentic Pipeline

Scaler School of Technology | ML System Design and LLMOps | Week 8 — Session 1/2 | Student Worksheet

The ML System Design Rubric — Your Framework for Today

Use this 10-step framework to design the system. Work through the steps in order on pages 2–3.

1. **Clarify the Problem** — scope, constraints, success metrics
2. **Context** — business objective, users, cost of wrong action
3. **Problem Statement** — input, output, what the agent decides
4. **Define Requirements** — functional and non-functional
5. **Estimations** — concurrent sessions, tokens/hour, LLM cost, tool call volume
6. **System Architecture** — (a) State management • (b) Data Engineering • (c) Feature Engineering • (d) Intent Routing • (e) Models • (f) Safety & Audit
7. **Evaluation** — offline metrics, online north-star, guardrails
8. **A/B Testing** — unit, split, duration, severity bias
9. **Deployment & Monitoring** — stateless pods, circuit breakers, cost ceiling
10. **Risk & Trade-offs** — loops, cost blow-up, prompt injection, when NOT to use agents

The Problem

Design a production AI customer support agent for an e-commerce platform.

Scale: 2 million customers, 10,000 concurrent conversations at peak.

Requirements: p99 latency < 5s per turn • handles order status, returns, product questions, complaint escalation • cost ceiling \$0.05 per resolved conversation • audit trail for every action • human escalation within 60s • graceful handling of prompt injection

Steps 1–3 — Clarify, Context, Problem Statement

Ask your clarifying questions, note the context, then write a precise problem statement.

Clarifying questions I would ask:

Context (business objective, users, cost of a wrong action):

Problem statement (input, output, what the agent decides at each step):

Step 4 — Requirements

Requirement	Value / assumption	Implication for design
Latency SLA per turn		
Concurrent sessions		
Cost ceiling		
Autonomous action scope		
Audit trail		

Step 5 — Estimations

What	Your estimate	Design implication
LLM tokens per hour		
LLM cost per hour		
Tool API calls per minute		
Session state size in Redis		

Step 6 — System Architecture

Label each component. CRITICAL: where do you enforce max-turn limit and cost ceiling?

(a–b) State Management & Data Engineering

(c–d) Customer Context & Intent Routing

(e–f) Models, Tools & Safety + Full Agent Flow Diagram

Draw the agent loop: classify → route → tool call → LLM → response. Mark where loop detection and cost ceiling fire.

Step 7 — Evaluation

Type	Metric(s)	Why this metric?
Offline		
Online north-star		
Guardrail		

Step 8 — A/B Testing

Unit of randomisation, split, duration, and the severity bias specific to customer support.

Step 9 — Deployment and Monitoring

Deployment (stateless pods, circuit breakers, hard limits):

8 metrics to monitor:

1.
2.
3.
4.
5.
6.
7.
8.

Step 10 — Risk Management and Trade-offs

Trade-off	Your choice	Reasoning
Autonomous vs. human approval for high-stakes actions		
Single LLM vs. routing to specialised models		
Full history vs. sliding window context		
Loop prevention strategy		
When NOT to use an agent here		

An Angry Customer Sends 50 Messages in 10 Minutes. What Happens?

Trace through your system: what fires, what limits apply, and how does the customer end up with a human agent?

Name: _____ Group: _____ Date: _____